

Machine Learning

1. What is Machine Learning

Machine learning is the study of computer programs that improve their performance on a task through experience, without being explicitly programmed for every possible input. A learning algorithm takes a set of examples and produces a model that can make predictions on new, unseen data.

The central idea is that the program learns a function from input features to an output value. The quality of the learned function depends on the amount and quality of the training data, the choice of the model family, and how the training procedure is configured.

A model is a concrete learned object, such as a decision tree or a set of weights, that the algorithm produces after training. The algorithm is the procedure used to find that model from the data. The data is organized as rows, where each row is one example, and columns, where each column is one attribute or feature of the example.

Machine learning systems appear throughout everyday technology. Search engines rank results, email services filter spam, streaming services recommend shows, and virtual assistants transcribe speech. What unifies these examples is that the behaviour is learned from data rather than written by hand. This makes machine learning especially useful for problems where it is hard to write an exact set of rules by hand, such as recognising objects in images or understanding free-form language.

2. Supervised and Unsupervised Learning

In supervised learning, every training example comes with a label. The algorithm learns to map features to those known labels. Classification is the supervised task where the output is one of a finite set of categories, such as deciding whether an email is spam or not spam.

Regression is the supervised task where the output is a continuous number, such as predicting the price of a house from its size and location. In both cases the labels are known during training, which is why the learning is described as supervised: a teacher provides the correct answers that the algorithm is trying to reproduce.

In unsupervised learning, the training data has no labels. The algorithm must find structure on its own. Clustering groups similar examples together, such as grouping customers by purchasing behaviour. Dimensionality reduction compresses the data into fewer dimensions while trying to preserve the important relationships between examples.

A third family is reinforcement learning, where an agent learns by taking actions in an environment and receiving rewards or penalties for the outcomes of those actions. The agent's goal is to choose a policy that maximises the total reward it receives over time, rather than trying to reproduce fixed labels from a dataset.

3. Training, Testing and Generalization

A dataset is usually split into a training set and a test set. The training set is used to fit the model, and the test set is used to estimate how well the model performs on data it has never seen. This measures generalization, which is the ability of the model to make useful predictions on new examples it did not encounter during training.

A model that memorizes the training data but fails on new data is said to overfit. Overfitting happens when the model is too complex relative to the amount of training data, so it learns noise rather than the underlying pattern. A complex model has enough capacity to match every detail of the training set, including random fluctuations that do not generalise.

Underfitting is the opposite problem, when the model is too simple to capture the pattern at all. A very flexible model that is allowed to become arbitrarily complex can fit the training data perfectly but loses its ability to generalise, while a very rigid model may never capture the pattern even in the training data.

The goal is to balance model complexity so the model generalizes well to unseen examples. Techniques such as cross-validation and regularisation help find this balance. Cross-validation repeats the training and evaluation process on several different splits of the data, and regularisation adds a penalty for excessive complexity so that the model prefers simpler explanations when the data does not justify more detail.

4. Features and Evaluation

Features are the individual measurable properties used as input to the model. Feature engineering is the process of selecting and transforming features so that they are informative for the task.

Good features capture the structure that the model needs to make predictions. Sometimes raw data is not directly usable, so it must be cleaned, normalised, or combined into derived features. Removing irrelevant features can also reduce overfitting by simplifying the model.

Accuracy is the fraction of test examples that the model labels correctly. Precision measures how many of the positively predicted examples are actually correct. Recall measures how many of the true positive examples the model managed to find.

There is a trade-off between precision and recall. Raising the bar for what counts as a positive prediction tends to increase precision but decrease recall, and lowering it does the opposite. For an email spam filter a false positive could mean deleting an important message, so precision matters most, whereas for cancer screening a missed case is far worse, so recall matters most.